

Early Identification of Heart Attack vs Gastritis
Using AI/ML
R26-IT-012

Project Proposal Report

Peiris.B.A– IT22193490

B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

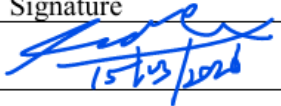
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Peiris.B.A	IT22193490	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
SMB Harshanath	15/03/2026	

ABSTRACT

This research proposes an AI-assisted diagnostic workflow for the early differentiation of heart attack (myocardial infarction) and gastritis-related chest pain, addressing a critical clinical problem where overlapping symptoms often lead to misinterpretation and delayed treatment. The proposal identifies that most existing AI-based diagnostic systems focus on a single condition, especially cardiovascular disease, and rarely support early differential diagnosis between cardiac and gastric causes of chest pain. It also highlights the lack of integrated educational decision-support tools for medical students and junior clinicians. To address this gap, the proposed system introduces a multi-stage diagnostic framework that begins with patient screening using demographic data, medical history, lifestyle factors, and symptom characteristics. In the first stage, a baseline risk assessment component applies dual-risk scoring and machine learning models such as XGBoost and LightGBM to estimate cardiac risk probability and gastric symptom probability. Borderline or high-risk cases are then validated through ECG and Troponin analysis, followed by cardiac imaging or gastric analysis depending on the pathway selected. The system is designed using a privacy-preserving approach by relying only on anonymized clinical attributes and public medical datasets. Expected outcomes include improved early screening accuracy, reduced risk of misdiagnosis, structured clinical decision support, and enhanced learning of differential diagnosis for medical students. In addition to its healthcare value, the proposed solution shows commercialization potential in medical education institutions, hospitals, and training environments through an AI-based clinical learning and decision-support platform.

Keywords: Artificial Intelligence, Chest Pain Diagnosis, Myocardial Infarction, Gastritis, Machine Learning, Machine Learning

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our supervisor, Mr. Buddika Harshanath, for his continuous guidance, valuable feedback, and encouragement throughout the development of this research proposal. His insightful suggestions and academic support greatly contributed to shaping the direction and quality of this research work.

We are also grateful to our co-supervisor, Prof. Samantha Rajapaksha, for his valuable advice, constructive comments, and continuous support during the planning and preparation of this research project.

We would like to extend our sincere appreciation to our external supervisors, Dr. Nishan Vas Gunawardhana and Dr. Hasintha from the Teaching Hospital – Polonnaruwa, for providing valuable clinical insights and professional guidance related to the medical aspects of this research. Their expertise and suggestions were extremely helpful in understanding the clinical perspective of chest pain diagnosis.

Our sincere thanks also go to the Sri Lanka Institute of Information Technology (SLIIT) and the Department of Information Technology for providing the academic environment, facilities, and resources required to carry out this research.

Finally, we would like to express our heartfelt gratitude to our parents, friends, and colleagues for their continuous support, encouragement, and motivation throughout our academic journey and during the completion of this research proposal.

TABLE OF CONTENT

Table of Contents

DECLARATION	2
ABSTRACT.....	3
ACKNOWLEDGEMENT	4
TABLE OF CONTENT	5
LIST OF TABLES	7
LIST OF FIGURES	8
LIST OF ABBREVIATIONS.....	9
LIST OF APPENDICES	10
INTRODUCTION	10
1.1 Background and Context.....	10
1.2 Literature Review.....	12
1.3 Research Gap	14
2. OBJECTIVE	17
2.1 Main Objective.....	17
2.2 Specific Objectives	17
3. METHODOLOGY	18
3.1 Research Approach	18
3.2 Technical Methods and Algorithms	19
3.3 Tools and Platforms with justification	19
3.4 Data Collection and Ethical Considerations	20
3.5 Validation Strategy and Performance Metrics	20
4. HIGH-LEVEL SYSTEM ARCHITECTURE	21
4.1 Overview of the Proposed Solution	21
4.2 Overall System Diagram.....	22
4.3 Highlight: Individual Component and Its Parts	22
4.4 Component Interactions	25
4.6 Data Flow Explanation	25
4.7 Scalability and Security Considerations	26
5. USER REQUIREMENTS	27
5.1 Stakeholder Identification.....	27
5.2 Functional Requirements	27

5.3 Non- Functional Requirements	28
6.COMMERCIALIZATION PLAN	29
6.1 Target Market and Customer Persona	29
6.2 Value Proposition	30
6.3 Revenue Model	30
6.4 Cost Estimation.....	31
Total Estimated Cost: ~100,000 LKR.....	31
6.5 Pricing Strategy.....	31
6.6 Competitive Advantage.....	32
6.7 Intellectual Property Considerations.....	32
7. BUDGET AND JUSTIFICATION.....	33
7.1 Estimated Budget Breakdown.....	33
7.2 Hardware Cost Justification	34
7.3 Software Requirements	34
7.4 Cloud Services	34
7.5 Human Effort Cost	35
8. WORK BREAKDOWN STRUCTURE (WBS).....	36
8.1 Task Breakdown per Member	36
8.2 Timeline Alignment	37
8.3 Balanced Workload Distribution.....	38
9. GANTT CHART	39
References.....	39
11. APPENDICES	41

LIST OF TABLES

Table 1 :Research Gap Matrix-Comparison of Existing Studies and Proposed System.....	17
Table 2: Functional Requirements for Component 1 – Baseline Risk & Patient Profile Analysis.....	28
Table 3: Value Proposition	30
Table 4: Estimated Development Cost for the Proposed AI Diagnostic Component	31
Table 5: Overall Budget Breakdown for the Proposed AI Diagnostic System Total Estimated Budget: 100,000 LKR.....	33
Table 6: Software Requirements for System Development.....	34
Table 7: Cloud Infrastructure Cost.....	35
Table 8: Human Effort and Research Activity Cost Estimation	36
Table 9: Work Breakdown Structure (WBS) – Task Breakdown for Component 1 Development.....	37
Table 10: Project Development Timeline for Component 1 – Baseline Risk & Patient Profile Analysis.....	38

LIST OF FIGURES

Figure 1: Integrated Overview	22
Figure 2: Patient Screening and Risk Assessment Workflow of Component 1	22
Figure 3: Data flow of Diagnostic System.....	26
Figure 4: Intellectual Property Protection Strategy for the AI Diagnostic Decision Support System.....	33
Figure 5: Hierarchical Work Breakdown Structure for Component 1 – Baseline Risk & Patient Profile Analysis.....	38
Figure 6: Gantt chart	39

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
ECG	Electrocardiogram
CVD	Cardiovascular Disease
MI	Myocardial Infarction
CNN	Convolutional Neural Network
1D-CNN	One-Dimensional Convolutional Neural Network
3D-CNN	Three-Dimensional Convolutional Neural Network
XAI	Explainable Artificial Intelligence
SHAP	Shapley Additive Explanations
API	Application Programming Interface
DSR	Design Science Research
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
F1-Score	Harmonic Mean of Precision and Recall
PCI	Percutaneous Coronary Intervention
CABG	Coronary Artery Bypass Grafting
PTB-XL	Physikalisch-Technische Bundesanstalt ECG Dataset
GI	Gastrointestinal
SaaS	Software as a Service
SDG	Sustainable Development Goal
SQL	Structured Query Language
PII	Personally Identifiable Information

LIST OF APPENDICES

Appendix	Title
Appendix A	Survey Instruments
Appendix B	Preliminary UI mockups
Appendix C	Risk Analysis
Appendix D	Ethical Documents

INTRODUCTION

1.1 Background and Context

Cardiovascular diseases (CVDs) are one of the leading causes of death worldwide and continue to pose a significant burden on global healthcare systems. According to global health reports, cardiovascular diseases account for millions of deaths each year and remain a major contributor to premature mortality. Among these conditions, myocardial infarction (commonly known as a heart attack) is a critical medical emergency that requires rapid identification and timely medical intervention. However, the early diagnosis of myocardial infarction remains challenging because its symptoms frequently overlap with those of other non-cardiac conditions, particularly gastrointestinal disorders such as gastritis, heartburn, or acid reflux [1] , [4]

Chest pain is the most common symptom associated with myocardial infarction. However, similar chest discomfort can also occur due to gastrointestinal problems, making it difficult for patients and even healthcare providers to correctly interpret the cause during the early stages of symptom presentation. Research has shown that many individuals mistakenly interpret heart attack symptoms as gastric problems, especially when symptoms include epigastric pain, nausea, or burning sensations in the chest [1]. This misconception may lead patients to delay seeking urgent medical care, which can significantly increase the risk of complications and mortality. Several clinical case studies have reported situations where myocardial infarction initially presented as gastrointestinal symptoms leading to delayed diagnosis or misinterpretation of the condition. [4], [5]

The impact of such misinterpretation is particularly significant because early treatment is critical in improving survival rates for cardiac patients. In many healthcare settings, the early assessment of chest pain depends heavily on clinical judgment and patient symptom evaluation before confirmatory tests are performed. Traditional diagnostic methods rely on medical tests such as electrocardiograms (ECG), cardiac biomarkers such as troponin levels, echocardiography, and coronary angiography to confirm cardiac events. While these tests are highly effective in confirming myocardial infarction, they are typically performed only after clinicians suspect a cardiovascular condition [9] .

Recent advancements in artificial intelligence and machine learning have introduced new opportunities for improving medical diagnostics. Machine learning models have been widely used for predicting cardiovascular diseases based on patient medical data, risk factors, and

clinical indicators [8] , [10]. Deep learning techniques have also been applied to analyze ECG signals and detect myocardial infarction automatically with high accuracy [9]. These approaches have significantly improved the capability of computer-assisted diagnosis in cardiology.

In addition to disease prediction models, digital healthcare technologies have also contributed to improving healthcare monitoring and medical data analysis. Mobile health systems enable remote monitoring of cardiac patients and improve healthcare service delivery through digital platforms [3] . Similarly, predictive analytics techniques have been used to enhance healthcare decision-making and optimize healthcare operations using large-scale medical data [2]. Educational technologies such as mixed-reality learning environments have also been developed to improve the training of medical students by providing interactive visualization of cardiovascular diseases and anatomical structures [7].

Despite these advancements, existing research reveals several limitations in current diagnostic systems. Most artificial intelligence models are designed to detect or predict a single medical condition, particularly cardiovascular diseases. However, in real clinical scenarios, chest pain can originate from multiple causes, including both cardiac and gastrointestinal conditions. Current systems rarely provide integrated diagnostic frameworks that assist clinicians in differentiating between cardiac and gastric causes of chest pain during the early stages of patient assessment. Furthermore, many machine learning models rely on a single type of medical data, such as ECG signals or patient clinical records, without integrating multiple diagnostic indicators into a unified decision-making workflow [11] [15].

Another limitation is the lack of educational decision-support systems that guide medical students or junior clinicians through the process of differential diagnosis. Although some technologies have been developed to support healthcare education, these systems mainly focus on visualization or theoretical learning rather than assisting users in practical diagnostic reasoning [7]. As a result, there is a need for intelligent systems that can support both clinical decision-making and medical education by guiding users through a structured diagnostic workflow.

To address these challenges, the proposed research introduces an AI-assisted diagnostic workflow designed to differentiate cardiac and gastric causes of chest pain during early patient screening. The proposed system integrates patient demographic information, medical history, lifestyle factors, and symptom characteristics to perform baseline risk assessment using machine learning techniques. In the first stage of the workflow, the system performs patient profile analysis and risk evaluation to estimate both cardiac risk probability and gastric symptom probability. This dual-risk evaluation approach enables early identification of patients who may require urgent cardiac assessment.

The proposed framework further integrates multiple diagnostic stages including electrocardiogram analysis, biomarker evaluation, and medical imaging analysis to support comprehensive clinical decision-making. By combining multiple sources of clinical information into a unified diagnostic workflow, the system aims to assist healthcare practitioners and medical students in improving diagnostic accuracy and reducing the risk of misinterpretation of chest pain symptoms.

From a global development perspective, the proposed research aligns with the United Nations Sustainable Development Goal 3 (SDG 3): Good Health and Well-Being, which aims to reduce

premature mortality from non-communicable diseases such as cardiovascular diseases through improved healthcare technologies and early detection mechanisms. Additionally, the integration of artificial intelligence technologies in healthcare supports SDG 9: Industry, Innovation, and Infrastructure, which promotes technological innovation and advanced digital solutions for improving healthcare systems.

From a research perspective, this study falls within the Artificial Intelligence and Data Science research cluster, particularly within the fields of medical informatics, machine learning, and healthcare analytics. The research combines clinical knowledge with modern artificial intelligence techniques to develop a decision-support system capable of assisting early diagnostic evaluation and improving healthcare decision-making processes.

Overall, the proposed research aims to address the limitations of existing diagnostic systems by developing an integrated AI-based diagnostic framework capable of assisting in the early differentiation between cardiac and gastric causes of chest pain. By providing structured decision support and integrating multiple diagnostic indicators, the system has the potential to improve early screening, support medical education, and enhance the effectiveness of AI-assisted healthcare technologies.

1.2 Literature Review

Recent advancements in healthcare analytics and artificial intelligence have significantly improved the capability of computer-assisted diagnostic systems, particularly in cardiovascular medicine. Numerous studies have explored the use of machine learning, deep learning, and healthcare data analytics to predict cardiovascular diseases, detect myocardial infarction, and support clinical decision-making processes. However, despite these advancements, several limitations remain in the early differential diagnosis of chest pain conditions, particularly in distinguishing cardiac causes from gastrointestinal disorders.

Several clinical studies have investigated the confusion between cardiac chest pain and gastric-related symptoms. Research comparing misconceptions of heart attack pain and heartburn symptoms found that a considerable number of individuals incorrectly interpret cardiac pain as gastric discomfort, especially when symptoms include burning sensations in the chest or upper abdominal pain [1]. Such misinterpretations may delay medical treatment and increase the risk of severe cardiac complications. Similar findings have been reported in clinical case studies where myocardial infarction initially presented with gastrointestinal symptoms such as nausea, vomiting, and epigastric discomfort, leading to diagnostic confusion and delayed intervention [4]. Additionally, studies of stress-induced cardiomyopathy, such as Takotsubo cardiomyopathy, have also highlighted the complex relationship between gastrointestinal conditions and cardiac symptoms, further emphasizing the need for improved diagnostic differentiation mechanisms [5].

With the increasing availability of healthcare data, machine learning techniques have been widely used for predicting cardiovascular diseases and assisting clinicians in identifying high-risk patients. Haq et al. proposed a hybrid machine learning framework that integrates multiple algorithms to improve heart prediction accuracy using clinical datasets [8]. Similarly, other studies have explored the use of machine learning models and predictive analytics techniques for analyzing healthcare datasets to improve medical decision-making and operational efficiency in healthcare systems [2]. These studies demonstrate the potential of artificial intelligence techniques to analyze complex medical data and support healthcare diagnostics.

Deep learning techniques have also been applied in cardiovascular diagnostics, particularly in analyzing electrocardiogram (ECG) signals for detecting myocardial infarction. allharya et al. developed a deep convolutional neural network model capable of automatically detecting myocardial infarction from ECG signals with high diagnostic accuracy [9]. Such deep learning approaches have significantly improved automated cardiac detection capabilities and reduced reliance on manual ECG interpretation. Furthermore, studies focusing on ECG data analysis have demonstrated the effectiveness of advanced signal processing and machine learning techniques in identifying abnormal cardiac patterns associated with heart diseases [22].

In addition to predictive models, clinical decision-support systems have been proposed to assist healthcare professionals in diagnosing cardiovascular conditions. Machine learning-based clinical decision-support frameworks have been developed to analyze patient medical records and risk factors to predict cardiovascular disease risk [10]. Similarly, intelligent decision-support systems using data mining techniques have been proposed to improve the accuracy of heart disease prediction by analyzing patient demographic information, clinical attributes, and lifestyle factors [13]. These systems demonstrate the potential of intelligent algorithms in assisting clinical decision-making processes.

Healthcare technology research has also explored the integration of digital healthcare systems and remote monitoring platforms to improve healthcare delivery. Mobile health intervention systems enable remote monitoring of cardiac patients and provide digital platforms for managing patient health data and communication between patients and healthcare providers [3]. Such systems contribute to improving healthcare accessibility and monitoring capabilities, particularly for patients with chronic cardiovascular conditions.

In addition to diagnostic and monitoring systems, educational technologies have also been introduced to support medical training. For example, mixed-reality learning environments have been developed to provide immersive learning experiences for medical students studying cardiovascular conditions. Systems such as Vascular Vision enable students to visualize cardiovascular anatomy and disease mechanisms in interactive environments, thereby enhancing the learning experience and improving conceptual understanding [7]. Although these technologies are useful for medical education, they mainly focus on visualization and learning rather than supporting practical diagnostic decision-making processes.

Recent literature also highlights the increasing role of artificial intelligence in cardiovascular medicine. Several studies have explored how artificial intelligence techniques can support precision cardiovascular medicine by analyzing large-scale healthcare datasets and identifying patterns associated with cardiovascular diseases [16], [18]. Big data analytics and machine learning technologies have further enabled the analysis of large biomedical datasets to improve diagnostic accuracy and clinical predictions [19]. These technological advancements demonstrate the growing importance of AI-based systems in modern healthcare environments.

Despite these advancements, several limitations remain in current diagnostic systems. Most existing machine learning models focus primarily on predicting or detecting a single medical condition, particularly cardiovascular diseases. However, chest pain symptoms can originate from multiple medical conditions, including gastrointestinal disorders such as gastritis or acid reflux. Current diagnostic systems rarely provide integrated frameworks that evaluate both cardiac and gastric causes simultaneously during the early stages of patient assessment.

Another major limitation of existing approaches is the lack of integrated diagnostic workflows that combine multiple sources of medical information. Many AI-based diagnostic systems analyze only a single type of data, such as ECG signals, clinical records, or medical imaging datasets. In contrast, real clinical decision-making requires physicians to consider multiple factors simultaneously, including patient symptoms, demographic information, medical history, and lifestyle factors. Research on chest pain risk stratification using machine learning has highlighted the need for integrated models capable of combining diverse clinical indicators to improve diagnostic accuracy in emergency departments [11], [15].

Furthermore, many existing healthcare technologies focus primarily on disease prediction or monitoring rather than supporting early differential diagnosis during patient screening. Artificial intelligence applications in healthcare are increasingly capable of predicting medical conditions, but fewer systems assist clinicians in evaluating multiple possible causes of symptoms during the early diagnostic stage. This limitation is particularly important in cases where patients present symptoms that may originate from different medical conditions.

Based on these observations, there is a clear need for integrated AI-assisted diagnostic systems capable of analyzing multiple clinical indicators and supporting differential diagnosis between cardiac and gastrointestinal causes of chest pain. The proposed research addresses this need by introducing a multi-stage diagnostic workflow that integrates patient symptom analysis, baseline risk assessment, electrocardiogram analysis, biomarker evaluation, and imaging-based assessments.

Component 1 of the proposed system focuses on baseline patient profiling and risk assessment, which includes analyzing demographic factors, medical history, lifestyle attributes, and symptom characteristics. This component acts as the initial screening stage of the diagnostic workflow and enables early evaluation of potential cardiac and gastric risk factors before advanced clinical tests are performed. By integrating multiple clinical indicators and machine learning techniques, the proposed system aims to provide a comprehensive diagnostic framework capable of supporting early clinical assessment and improving diagnostic accuracy.

Overall, the reviewed literature demonstrates that while significant progress has been made in cardiovascular disease prediction and healthcare AI technologies, existing systems rarely provide integrated diagnostic frameworks capable of differentiating cardiac and gastric causes of chest pain during early patient screenings. Therefore, the development of an AI-assisted diagnostic workflow that integrates symptom analysis, patient risk profiling, and multi-stage clinical evaluation represents an important contribution to improving early diagnosis and supporting medical education.

1.3 Research Gap

Despite significant advancements in artificial intelligence and machine learning applications in healthcare, several limitations remain in existing diagnostic systems, particularly in the early differentiation of chest pain conditions. Many current studies focus on predicting cardiovascular diseases using clinical datasets, electrocardiogram (ECG) signals, or medical imaging data. Machine learning frameworks have demonstrated strong performance in predicting heart disease risk and identifying cardiac abnormalities from medical records and clinical attributes [8], [10]. Similarly, deep learning techniques have been successfully applied to detect myocardial infarction through automated ECG signal analysis with high diagnostic

accuracy [9], [22]. Although these approaches significantly improve cardiovascular diagnostics, they mainly focus on identifying cardiac diseases rather than assisting clinicians in differentiating cardiac symptoms from other medical conditions during early patient assessment.

One of the major limitations identified in existing literature is the single condition focus of many artificial intelligence models. Most diagnostic systems are designed specifically to detect or predict heart diseases, even though chest pain can originate from multiple causes, including gastrointestinal disorders such as gastritis or acid reflux. Clinical research has shown that many patients mistakenly interpret heart attack symptoms as gastric discomfort, especially when symptoms include burning sensations in the chest or epigastric pain [1]. Several clinical case studies have also reported situations where myocardial infarction initially presented with gastrointestinal symptoms, which led to delayed diagnosis or misinterpretation of the underlying cardiac condition [4], [5]. Despite these observations, very few computational systems have been developed to assist clinicians in differentiating cardiac chest pain from gastrointestinal causes during the early screening stage.

Another important limitation of existing diagnostic technologies is the lack of early screening tools designed for medical students and junior clinicians. Most clinical decision-support systems are developed to confirm cardiovascular events using diagnostic tests such as ECG analysis, cardiac biomarkers, and medical imaging techniques [9], [10]. However, these systems are generally used only after a clinician suspects of a cardiac condition. In real clinical practice, the initial evaluation of chest pain typically depends on patient symptoms, medical history, and preliminary clinical judgment. This early diagnostic stage can be particularly challenging for medical students or less experienced healthcare practitioners who may struggle to interpret overlapping symptoms between cardiac and gastrointestinal conditions.

Furthermore, many existing machine learning models rely on single-source medical data, such as ECG signals or patient clinical records [8], [11], [15]. In contrast, real clinical diagnosis requires the integration of multiple sources of information, including patient demographics, lifestyle factors, medical history, symptom characteristics, and laboratory test results. The lack of integrated systems capable of combining these diverse clinical indicators limits the ability of current technologies to support comprehensive differential diagnosis.

In addition, several digital healthcare technologies focus primarily on healthcare monitoring or data management rather than supporting diagnostic reasoning. Mobile health platforms improve remote patient monitoring and healthcare delivery [3], while predictive analytics techniques enhance healthcare data analysis and system efficiency [2]. Educational technologies such as mixed-reality medical training systems also improve learning experience for medical students by visualizing cardiovascular anatomy and disease mechanisms [7]. However, these technologies rarely provide integrated diagnostic frameworks that guide users through the clinical reasoning process required for evaluating chest pain conditions.

To address these limitations, the proposed research introduces an AI-assisted diagnostic workflow designed to differentiate cardiac and gastric causes of chest pain during early patient screening. The proposed system introduces a dual-risk scoring mechanism that simultaneously

evaluates both cardiac risk probability and gastric symptom probability during the initial assessment stage. This approach enables early differentiation of potential causes of chest pain before advanced diagnostic tests are performed.

Additionally, the proposed research introduces a multi-stage diagnostic framework that integrates symptom analysis, patient profile assessment, ECG signal evaluation, biomarker analysis, and medical imaging analysis into a unified decision-support system. This workflow allows multiple clinical indicators to be evaluated sequentially, improving the reliability of early clinical assessment. The system is also designed as an educational diagnostic platform for medical students, enabling users to learn clinical reasoning and differential diagnosis through structured diagnostic workflows.

Overall, although existing research has made substantial progress in cardiovascular disease prediction and artificial intelligence-based healthcare technologies, integrated diagnostic systems capable of differentiating cardiac and gastric causes of chest pain during early patient screening remain limited. Therefore, the proposed research aims to develop an AI-assisted diagnostic framework that combines patient symptom analysis, baseline risk assessment, and multi-stage clinical evaluation to support early diagnostic decision-making and improve medical education.

Research Study	Heart Disease Prediction	Cardiac vs Gastric Differentiation	Early Screening Support	Multi-Data Integration	Educational Decision Support	Integrated Diagnostic Workflow
Pervez et al. – Heart Attack vs Heartburn Misconception Study [1]	✓	✓	✗	✗	✗	✗
Predictive Healthcare Analytics System [2]	✓	✗	✗	✗	✗	✗
Mobile Health Cardiac Monitoring System [3]	✓	✗	✓	✗	✗	✗
Atypical Myocardial Infarction Case Study [4]	✓	✓	✗	✗	✗	✗
Takotsubo Cardiomyopathy Clinical Study [5]	✓	✓	✗	✗	✗	✗
Blockchain Healthcare Monitoring System [6]	✗	✗	✗	✗	✗	✗
VascularVision Mixed Reality Learning System [7]	✗	✗	✗	✗	✓	✗
Machine Learning Heart Disease Prediction (Haq et al.) [8]	✓	✗	✗	✗	✗	✗
Deep Learning ECG Detection (Acharya et al.) [9]	✓	✗	✗	✗	✗	✗

Clinical Decision Support System for Heart Disease [10]	✓	X	X	X	X	X
Chest Pain Risk Prediction Models [11], [15]	✓	X	✓	X	X	X
AI-Based Cardiovascular Medicine Systems [16], [18], [19]	✓	X	X	X	X	X
Proposed AI-Assisted Diagnostic Workflow	✓	✓	✓	✓	✓	✓

Table 1 :Research Gap Matrix-Comparison of Existing Studies and Proposed System

Table 1 presents a comparison between existing research studies and the proposed AI-assisted diagnostic workflow. Most existing studies focus primarily on cardiovascular disease prediction using machine learning models or ECG-based analysis. Although several studies address chest pain assessment and cardiac risk prediction, very few systems support the differentiation of cardiac chest pain from gastrointestinal causes such as gastritis during the early screening stage. Additionally, many existing systems rely on a single type of medical data and do not integrate multiple diagnostic indicators into a unified clinical workflow. Furthermore, current technologies rarely provide educational decision-support platforms that guide medical students or junior clinicians through structured diagnostic reasoning.

The proposed system addresses these limitations by introducing an integrated AI-assisted diagnostic framework that combines baseline patient risk assessment, symptom analysis, multi-data integration, and a dual-risk scoring mechanism for cardiac and gastric conditions. This workflow supports early patient screening while also serving as an educational platform for medical students learning differential diagnosis of chest pain conditions.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop an AI-assisted diagnostic workflow that supports early differentiation between cardiac and gastric causes of chest pain by analyzing patient symptoms, medical history, and clinical indicators, thereby improving early screening accuracy and assisting medical students in understanding differential diagnosis.

This system integrates machine learning techniques, patient risk analysis, and multi-stage diagnostic evaluation to support clinical reasoning and early decision-making before advanced medical tests are performed.

2.2 Specific Objectives

To achieve the main objective, the following specific objectives are defined:

- **Develop a patient profiling and baseline risk assessment module**
To design and implement a system that collects patient demographic information,

lifestyle factors, medical history, and symptom characteristics in order to perform initial risk analysis for cardiac and gastric conditions.

- **Implement a dual-risk scoring mechanism for early screening**
To develop a computational model that simultaneously evaluates **cardiac risk probability and gastric symptom probability**, enabling early differentiation of potential causes of chest pain before advanced clinical tests are conducted.
- **Integrate artificial intelligence techniques for clinical data analysis**
To apply machine learning algorithms to analyze patient clinical attributes and support intelligent decision-making in identifying high-risk patients requiring further cardiac evaluation.
- **Design a multi-stage diagnostic workflow for chest pain assessment**
To develop an integrated workflow that connects **patient screening, risk assessment, ECG analysis, biomarker evaluation, and imaging analysis** to guide users through a structured diagnostic process.
- **Develop a decision-support platform for medical education**
To create an AI-assisted diagnostic platform that supports medical students and junior clinicians in learning clinical reasoning and differential diagnosis between cardiac and gastrointestinal conditions.
- **Evaluate the effectiveness of the proposed diagnostic system**
To assess the performance and usability of the proposed system in supporting early diagnostic evaluation and improving understanding of chest pain assessment among medical students.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) approach combined with experimental evaluation. The design science methodology is appropriate because the study aims to develop an AI-assisted diagnostic support system that solves a real-world clinical problem.

The proposed system is designed to support early differentiation between cardiac-related chest pain and gastric-related symptoms using patient clinical attributes and machine learning techniques. The research involves designing, implementing, and evaluating a prototype system that integrates multiple diagnostic stages including patient screening, risk scoring, and validation tests.

Within the overall system workflow, Component 1 focuses on patient profile analysis and baseline risk assessment, which acts as the initial screening layer before advanced diagnostic procedures such as ECG analysis and imaging tests are performed. The system collects patient questionnaire data and calculates preliminary cardiac and gastric risk scores to guide further diagnostic pathways.

The experimental phase evaluates the accuracy and reliability of the AI models using structured medical datasets and simulated patient cases.

3.2 Technical Methods and Algorithms

The technical implementation of Component 1 combines rule-based clinical scoring and machine learning techniques to evaluate early risk indicators.

The system uses a dual risk scoring mechanism which calculates two independent values:

- Cardiac Score – representing the probability of cardiac-related chest pain
- Gastric Score – representing the probability of gastrointestinal causes

These scores are calculated using patient clinical attributes such as:

- pain duration and location
- nausea and sweating
- aggravating and relieving factors
- risk factors (diabetes, hypertension, smoking)
- age and gender
- past ischemic heart disease

Based on these scores, the system classifies patients into three preliminary categories:

- Preliminary Cardiac
- Preliminary Non-Cardiac (Gastric)
- Borderline (requires ECG and Troponin testing)

This scoring approach follows clinical triage logic and reduces the risk of misinterpreting cardiac symptoms such as gastritis.

To improve prediction performance, machine learning algorithms such as XGBoost and LightGBM are applied to analyze structured clinical data. These gradient boosting models are suitable for tabular medical datasets and can capture complex relationships between risk factors. Additionally, explainable AI techniques are incorporated to identify the most influential variables contributing to risk prediction.

3.3 Tools and Platforms with justification

The proposed system uses a modern AI-enabled development stack designed to support machine learning, medical data processing, and web-based interaction.

- **Programming Language – Python**
Python is used as the primary programming language because it provides extensive support for machine learning libraries, medical data analysis, and rapid prototyping.
- **Backend Framework – FastAPI**
FastAPI is selected for developing backend APIs that integrate AI models with the application interface. It enables efficient communication between the machine learning modules and the user interface.
- **Machine Learning Libraries – Scikit-learn, XGBoost, LightGBM**
These libraries are used to build predictive models for patient risk assessment.

Gradient boosting algorithms are particularly effective for structured questionnaire data and provide high prediction accuracy.

- **Data Processing Tools – NumPy and Pandas**

These tools support data preprocessing, feature engineering, and statistical analysis of patient datasets.

- **Database – PostgreSQL**

PostgreSQL is used to store questionnaire responses, patient attributes, and prediction results because it provides reliable structured data management and strong data integrity.

- **Explainable AI Tools – SHAP**

SHAP (Shapley Additive explanations) is used to interpret model predictions by identifying which patient attributes contribute most to cardiac or gastric risk classification.

- **Frontend Framework – React.js**

React is used to develop an interactive dashboard where medical students can enter patient data, visualize risk scores, and observe AI-assisted diagnostic pathways.

This technology stack enables efficient integration of machine learning models, data processing, and interactive clinical visualization.

3.4 Data Collection and Ethical Considerations

The system follows a privacy-preserving data collection approach to ensure that patient confidentiality is maintained.

The system does not collect any personally identifiable information such as patient name, national identification number, address, or phone number. Instead, it only collects anonymized clinical attributes including age range, gender, symptoms, lifestyle factors, and medical history.

Training data for machine learning models is obtained from publicly available medical datasets such as ECG and gastrointestinal imaging datasets, which are already anonymized and widely used in academic research.

Additionally, synthetic patient profiles can be generated to simulate different clinical scenarios for training and testing purposes. This approach ensures that the research complies with ethical guidelines and data protection principles while still enabling effective model development.

3.5 Validation Strategy and Performance Metrics

The effectiveness of the proposed system is evaluated using experimental validation techniques.

The dataset is divided into training and testing subsets to evaluate the predictive performance of the machine learning models. Cross-validation techniques are applied to ensure that the models generalize well across different patient profiles.

The performance of the models is assessed using standard classification metrics including:

- **Accuracy** – overall correctness of predictions
- **Precision** – ability to correctly identify cardiac risk cases
- **Recall (Sensitivity)** – ability to detect true cardiac events
- **F1-Score** – balance between precision and recall
- **ROC-AUC** – model discrimination ability

In addition to quantitative evaluation, usability testing will be conducted to assess how effectively the system supports medical students in understanding the diagnostic workflow and interpreting AI predictions.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed research system is an AI-assisted clinical decision support platform designed to differentiate cardiac-related chest pain from gastric-related chest pain through a structured diagnostic workflow. The complete solution consists of four major components that work together to simulate real clinical reasoning and support medical student learning.

Component 1 – Baseline Risk & Patient Profile Analysis performs the initial screening stage using questionnaire data, patient demographics, medical history, lifestyle factors, and early biomarkers. It calculates cardiac and gastric risk probabilities and categorizes the patient into the appropriate preliminary pathway.

Component 2 – Heart Signal & Functional Analysis processes ECG and echocardiography data to detect cardiac abnormalities. This component supports confirmation of suspected cardiac conditions through AI-assisted signal and image analysis.

Component 3 – Coronary Blockage Detection analyzes coronary angiography data to identify arterial blockage location, severity, and treatment direction. This component supports final cardiac pathway decision-making.

Component 4 – Gastric Analysis and Decision Support Engine is activated only when ECG and Troponin results indicate that the patient belongs to the gastric pathway. This module receives symptom triggers from Component 1 together with gastroscopy/endoscopy images, applies image preprocessing and EfficientNet-B4-based analysis, confirms the gastric disease, determines severity, and generates treatment recommendations. The component also supports medical students by providing diagnostic explanations and feedback.

4.2 Overall System Diagram

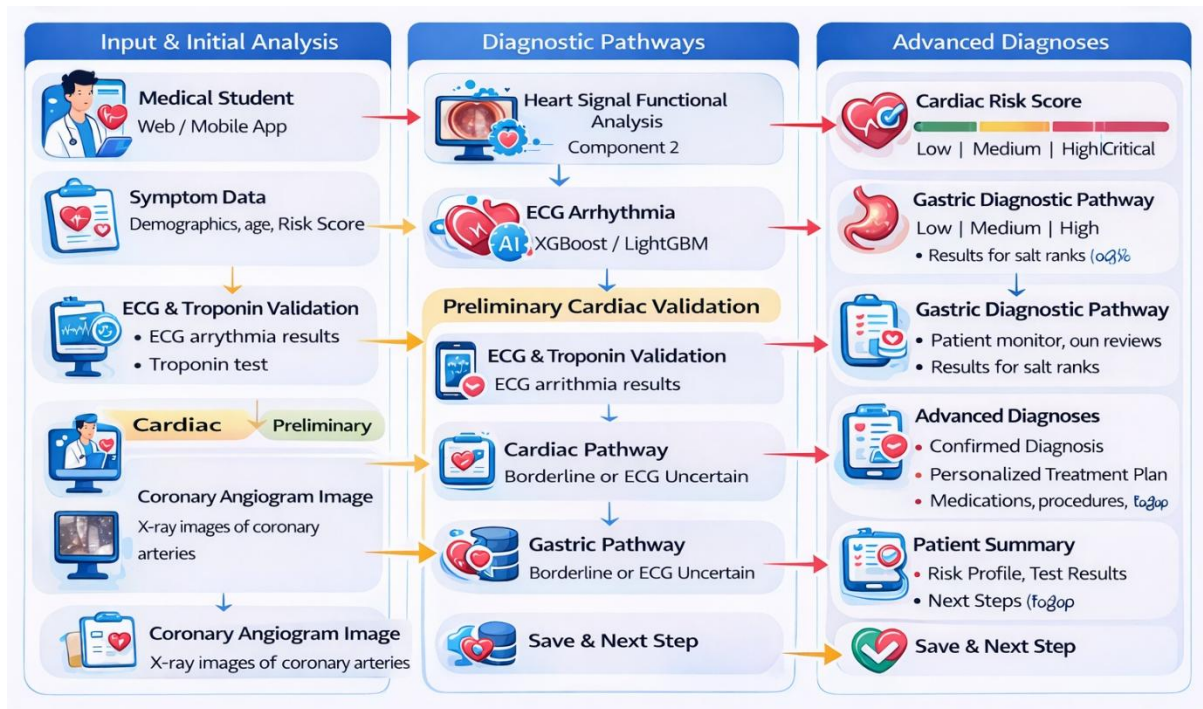


Figure 1: Integrated Overview

4.3 Highlight: Individual Component and Its Parts

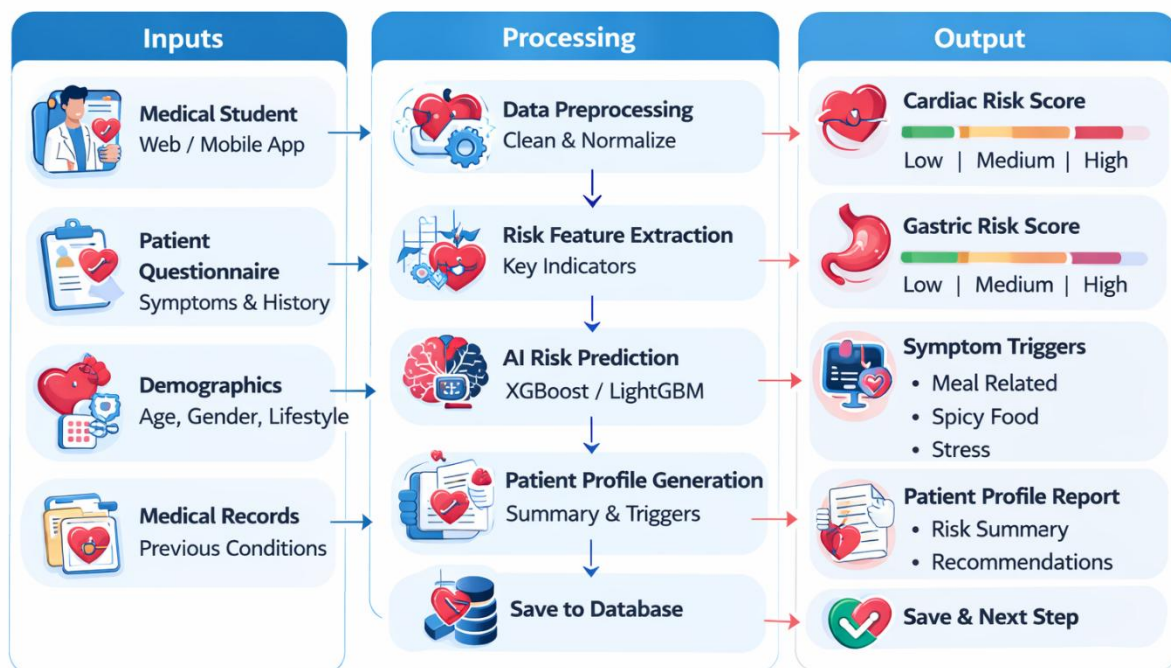


Figure 2: Patient Screening and Risk Assessment Workflow of Component 1

Component 1 represents the **initial patient screening and risk evaluation stage** of the proposed system. It serves as the **entry point of the diagnostic workflow**, where patient clinical information is collected and analyzed to estimate the likelihood of cardiac or gastric causes of chest pain. This component processes structured patient data using rule-based scoring and machine learning models to generate early risk predictions and guide the next diagnostic pathway.

The internal architecture of this component consists of the following modules:

1. Patient Data Input Module

This module collects patient information through a **web or mobile-based questionnaire interface** used by medical students or healthcare trainees. The data collected includes:

- **Demographic information:** age, gender
- **Lifestyle factors:** smoking habits, stress levels
- **Medical history:** diabetes, hypertension, previous cardiac conditions
- **Symptom characteristics:** pain duration, location, triggers, nausea, sweating
- **Previous medical conditions or medications**

This module ensures that the system receives structured clinical attributes required for early risk evaluation.

2. Data Preprocessing Module

The collected patient data is processed to ensure quality and consistency before analysis. This module performs:

- data cleaning and normalization
- handling of missing values
- standardization of clinical variables

The processed dataset is then transformed into structured features that can be used by machine learning algorithms for predictive analysis.

3. Risk Feature Extraction Module

In this stage, the system extracts important clinical indicators from the processed patient data. These indicators include:

- symptom severity patterns
- pain location and duration characteristics
- presence of associated symptoms such as nausea or sweating
- lifestyle and medical risk factors

The extracted features represent the **key variables used for calculating cardiac and gastric risk scores**.

4. AI Risk Prediction Module

This module applies machine learning algorithms such as **XGBoost and LightGBM** to analyze patient attributes and estimate the probability of cardiac or gastric causes of chest pain. These models are well suited for structured clinical datasets and can identify complex relationships between symptoms, lifestyle factors, and medical history.

The AI model outputs probability values representing the likelihood of:

- cardiac-related chest pain
- gastric-related symptoms

5. Patient Profile Generation Module

Based on the extracted features and AI predictions, the system generates a summarized **patient risk profile**. This profile includes:

- major symptom triggers (e.g., food intake, stress)
- contributing risk factors
- summarized clinical indicators

The patient profile helps users understand the reasoning behind the system's prediction and supports educational learning for medical students.

6. Risk Scoring and Output Generation Module

Finally, the system produces the diagnostic outputs based on the calculated scores and model predictions. The outputs include:

- **Cardiac Risk Score (Low / Medium / High)**
- **Gastric Risk Score (Low / Medium / High)**
- **Symptoms trigger indicators**
- **Patient risk summary and recommendations**

These results help determine the next diagnostic pathway in the system, guiding the patient toward either **cardiac diagnostic tests or gastric analysis procedures**.

7. Data Storage Module

All processed patient data, risk scores, and prediction results are securely stored in the system database for future analysis, monitoring, and educational review.

4.4 Component Interactions

The system components interact through a centralized backend service that coordinates data exchange between AI models and the user interface.

1. The patient enters clinical data through the web interface.
2. The backend system forwards the data to Component 1 for risk assessment.
3. Based on the risk classification, the system routes the patient to either the cardiac diagnostic pathway or the gastric diagnostic pathway.
4. Cardiac cases are processed through ECG and echocardiogram analysis (Component 2) and angiography evaluation (Component 3).
5. Non-cardiac cases are processed through gastric disease analysis (Component 4).
6. The outputs from these components are integrated to produce the final diagnosis and educational feedback.

4.6 Data Flow Explanation

The system data flow follows the clinical diagnostic workflow.

- **Step 1 – Patient Screening**
Patient demographic data, medical history, and symptoms are collected through a questionnaire.
- **Step 2 – Risk Categorization**
Component 1 calculates cardiac and gastric risk scores and assigns a preliminary diagnostic category.
- **Step 3 – Secondary Validation**
If cardiac risk is suspected, ECG and Troponin validation tests are performed to confirm potential myocardial injury.
- **Step 4 – Cardiac Pathway**
Cardiac patients undergo further analysis using echocardiography and coronary angiography imaging models.
- **Step 5 – Gastric Pathway**
If cardiac conditions are excluded, gastric symptoms and endoscopy images are analyzed to detect gastrointestinal diseases.
- **Step 6 – Final Output**
The system produces:
 - risk profile
 - diagnostic interpretation
 - treatment recommendations
 - Educational explanations for students

- This workflow ensures systematic clinical reasoning supported by AI analysis.

Figure 3: Data flow of Diagnostic System

The system architecture is designed to support scalability and secure handling of medical data.

The architecture supports modular AI components, allowing each module to be independently improved or scaled. The backend API enables integration with cloud infrastructure and containerized deployment using Docker, allowing the system to support larger datasets and multiple users.

To protect patient privacy, the system does not store personally identifiable information such as names, identification numbers, or contact details. Only anonymized medical attributes are used for analysis.

- encrypted database storage
- role-based access control
- anonymized datasets for model training
- secure API communication

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholders are individuals or groups who interact with the system, contribute to its development, or benefit from its outcomes. The proposed AI-assisted diagnostic system involves several stakeholders including medical students, patients, researchers, and developers.

Medical Students

Medical students are the primary users of the system. The platform is designed to help them understand the clinical reasoning process involved in differentiating cardiac chest pain from gastric-related symptoms. Students can input patient information, observe risk scores generated by the AI models, and learn how different symptoms and risk factors influence the diagnostic pathway.

Patients

Patients indirectly contribute to the system by providing symptom information and clinical attributes during the screening stage. The system analyzes anonymized patient data such as age, gender, symptoms, and medical history to calculate cardiac and gastric risk probabilities. Patients benefit from the system through improved early screening and identification of potential cardiac risks.

Researchers

Researchers are responsible for analyzing the system's performance, evaluating machine learning models, and improving diagnostic algorithms. They use anonymized datasets generated from system interactions to study prediction accuracy, model performance, and healthcare decision-support effectiveness.

Developers

Developers design, implement, and maintain the system. They are responsible for developing machine learning models, integrating system components, managing the backend APIs, and ensuring the system functions efficiently and securely.

5.2 Functional Requirements

Functional requirements describe the specific system functions that must be implemented to support early chest pain screening and diagnostic risk analysis.

Requirement ID	Functional Requirement Description
FR1	The system shall allow users to enter patient demographic information such as age and gender.

FR2	The system shall allow users to input patient symptoms including chest pain duration, location, nausea, sweating, and pain characteristics.
FR3	The system shall allow users to record patient medical history including diabetes, hypertension, smoking habits, and family history of heart disease.
FR4	The system shall process questionnaire responses and convert them into structured features suitable for machine learning analysis.
FR5	The system shall calculate a Cardiac Risk Score based on cardiac-related symptoms and clinical risk factors.
FR6	The system shall calculate a Gastric Risk Score based on gastrointestinal-related symptoms and triggers.
FR7	The system shall apply predefined clinical rules to detect high-risk cardiac patterns such as central chest pain with sweating.
FR8	The system shall allow users to input Troponin biomarker values when available.
FR9	The system shall interpret Troponin levels to support cardiac risk assessment.
FR10	The system shall classify the patient into one of the following preliminary categories: Preliminary Cardiac, Preliminary Gastric, or Borderline.
FR11	The system shall display the calculated risk scores and the resulting classification to the user.
FR12	The system shall provide recommendations for the next diagnostic step such as ECG testing or gastric evaluation.
FR13	The system shall highlight the key symptoms or risk factors that contributed to risk prediction using explainable AI techniques.
FR14	The system should store anonymized screening data for research and system improvement purposes.

Table 2: Functional Requirements for Component 1 – Baseline Risk & Patient Profile Analysis

5.3 Non- Functional Requirements

Non-functional requirements describe the quality attributes and operational characteristics that ensure the system operates efficiently and reliably.

Security: The system must protect medical screening data using secure authentication and controlled access mechanisms. Sensitive health information must be handled securely to prevent unauthorized access.

Privacy: The system must avoid storing personally identifiable patient information such as name, national identification number, phone number, or address. Only anonymized clinical attributes should be collected for analysis to maintain patient privacy.

Performance: The system should generate screening results quickly after patient data submission. Fast response time is important because this module represents the first stage of the diagnostic workflow.

Accuracy: The system must produce reliable predictions using validated scoring rules and machine learning models. High prediction accuracy is important to minimize the risk of misclassifying cardiac cases.

Usability: The system interface should be simple and user-friendly so that medical students and healthcare trainees can easily enter patient information and understand the results.

Explainability: The system should provide explanations for AI predictions by highlighting the symptoms or risk factors that most influenced the classification decision.

Reliability: The system must produce consistent results for the same input data and handle missing or incomplete inputs appropriately.

Scalability: The system architecture should allow future expansion, including integration with additional diagnostic components such as ECG analysis and medical imaging modules.

Maintainability: The system should allow developers to easily update machine learning models, scoring rules, and system components without significant redesign.

6.COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed system focuses on the healthcare and medical education sectors where early chest pain assessment and clinical learning tools are important.

Target Market: The primary target market includes medical education institutions, hospitals, and healthcare training centers that require decision-support tools to assist medical students and junior practitioners in understanding early diagnostic workflows.

Customer Persona

1. Medical Students
 - Age group: 20–28
 - Background: Undergraduate or postgraduate medical students

- Needs: Tools that help them understand diagnostic reasoning and clinical decision-making.
2. Junior Practitioners
 - Background: Intern doctors or early-career medical professionals
 - Needs: Systems that assist with initial patient screening and structured clinical reasoning.
 3. Medical Training Institutions
 - Universities and teaching hospitals that require AI-based medical learning tools to improve clinical education.

6.2 Value Proposition

User Problem	Proposed Solution	Value to the User
Medical students struggle to understand early differential diagnosis of chest pain	AI-based screening system that analyzes symptoms and risk factors	Improves understanding of cardiac vs gastric diagnosis
Junior practitioners may find it difficult to quickly assess chest pain cases	Automated risk scoring using CardiacScore and GastricScore	Provides structured clinical decision support
Lack of interactive learning tools in medical education	Interactive system showing diagnostic reasoning	Enhances clinical learning experience
Risk of missing early cardiac symptoms during screening	Rule-based detection of high-risk cardiac patterns	Help identify potential cardiac emergencies earlier
Difficulty interpreting laboratory results such as Troponin	Automated Troponin value interpretation	Simplifies clinical analysis and reduces confusion

Table 3: Value Proposition

6.3 Revenue Model

The commercialization strategy for the system may include multiple revenue streams.

Institutional Licensing

- Universities and medical schools can license the system for educational use.

Hospital Subscription Model

- Hospitals or training hospitals can subscribe to the system for clinical training purposes.

Software as a Service (SaaS)

- The system can be offered as a cloud-based platform where institutions pay a monthly or annual subscription.

Research Collaboration

- Partnerships with healthcare research organizations for clinical studies and system improvements.

6.4 Cost Estimation

Cost Item	Description	Estimated Cost (LKR)
Hardware Usage	Personal laptop and minor hardware resources used for AI model development and testing	25,000
Cloud Hosting	Cloud server required to deploy the backend system and machine learning models	30,000
Database Setup	PostgreSQL database configuration and storage management	5,000
Software Tools	Development tools, IDE plugins, and version control services	5,000
Data Processing	Dataset preparation, preprocessing, and model training experiments	15,000
System Testing	System debugging, validation experiments, and performance evaluation	10,000
Research Travel & Miscellaneous	Transportation and operational expenses for research activities and documentation	10,000

Table 4: Estimated Development Cost for the Proposed AI Diagnostic Component

Total Estimated Cost: ~100,000 LKR

6.5 Pricing Strategy

The pricing strategy will follow a subscription-based model suitable for educational institutions and healthcare organizations.

Educational License

- Universities and medical schools can purchase an institutional license for student use.

Hospital Subscription

- Hospitals can subscribe to the system annually for training and diagnostic support.

Tiered Pricing Model

Different pricing tiers can be introduced depending on:

- Number of users
- Features included

- Institutional size

This strategy ensures accessibility while allowing scalable commercial deployment.

6.6 Competitive Advantage

The proposed system provides several advantages compared to existing medical decision-support tools.

Dual Disease Screening

- Unlike many systems that focus only on heart disease prediction, this system differentiates between cardiac and gastric causes of chest pain.

Educational Focus

- Designed specifically for medical student learning and clinical training.

Explainable Decision Support

- Provides explanations for risk classification, helping users understand diagnostic reasoning.

Integrated Risk Scoring

- Combines symptom analysis, risk factor evaluation, and Troponin interpretation into a single screening workflow.

Modular Architecture

- Allows integration with future diagnostic modules such as ECG analysis or imaging systems.

6.7 Intellectual Property Considerations

The proposed AI-assisted diagnostic system introduces innovative elements related to **machine learning–based risk assessment, dual-risk scoring algorithms, and clinical decision-support workflows** for differentiating cardiac and gastric causes of chest pain. These novel approaches may qualify for intellectual property protection.

Potential intellectual property considerations include:

- **Patent protection** for the dual-risk scoring mechanism and AI-assisted diagnostic workflow used to differentiate cardiac and gastric conditions during early screening.
- **Copyright protection** for the developed software, including machine learning algorithms, model architecture, and the diagnostic decision-support platform.
- **Data protection and licensing** for trained machine learning models developed using anonymized medical datasets.

- **Potential collaboration** with healthcare technology companies, hospitals, or medical software providers for commercialization and system deployment.

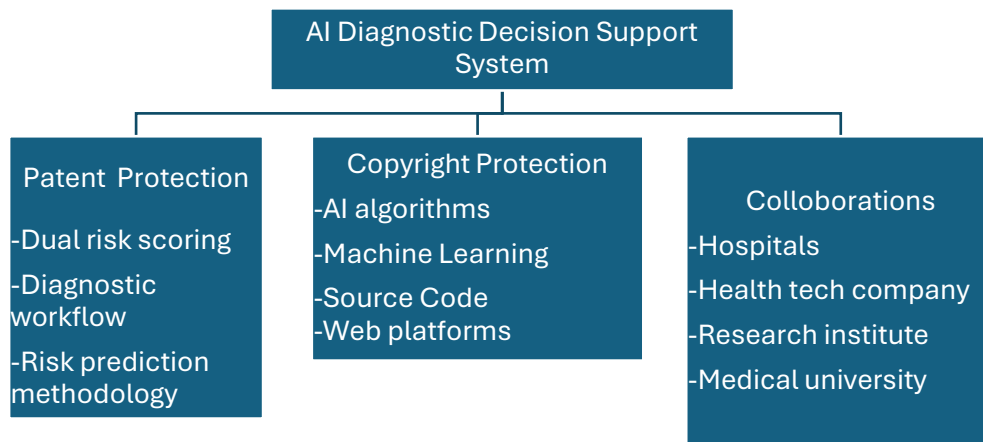


Figure 4: Intellectual Property Protection Strategy for the AI Diagnostic Decision Support System

7. BUDGET AND JUSTIFICATION

7.1 Estimated Budget Breakdown

Budget Category	Description	Estimated Cost (LKR)
Hardware	Personal laptop usage and minor hardware resources for development and testing	25,000
Software	Development tools, IDE support tools, and version control services	5,000
Cloud Services	Cloud server hosting for AI model deployment and testing	30,000
Data Processing	Dataset preparation, preprocessing, and experiment configuration	15,000
Human Effort	Development, system testing, and research implementation effort	20,000
Miscellaneous	Internet usage, documentation preparation, and research materials	5,000

Table 5: Overall Budget Breakdown for the Proposed AI Diagnostic System

Total Estimated Budget: 100,000 LKR

7.2 Hardware Cost Justification

The project requires computing resources to develop and train machine learning models for the baseline risk assessment component. The development activities include dataset preprocessing, training gradient boosting models, and testing the system locally.

Instead of purchasing new hardware, the project will utilize an existing personal laptop with sufficient processing power and memory. Therefore, only a small portion of the budget is allocated for hardware usage and potential upgrades such as additional storage devices or minor peripherals.

This approach significantly reduces development costs while still enabling efficient model training and system testing.

7.3 Software Requirements

Software Tool	Purpose	Cost
Python	Main programming language for AI model development	Free
Scikit-learn / XGBoost / LightGBM	Machine learning libraries for risk prediction models	Free
React.js	Development of the web-based user interface	Free
PostgreSQL	Database management for storing patient data	Free
Git / GitHub	Version control and collaboration	Free
Development Tools (IDE support)	Optional development plugins or tools	5,000

Table 6: Software Requirements for System Development

Justification: Most technologies used in the project are open-source tools widely adopted in artificial intelligence and web development. These technologies reduce licensing costs while providing powerful capabilities for building machine learning applications.

7.4 Cloud Services

Cloud Service	Purpose	Estimated Cost (LKR)
Cloud Server (AWS / GCP / Azure)	Hosting backend API and AI models	18,000
Cloud Storage	Storage for datasets and experiment results	7,000

Deployment Services	Container deployment and server configuration	5,000
---------------------	---	-------

Table 7: Cloud Infrastructure Cost

Justification: Cloud services enable remote deployment of the AI models and allow the system to be accessed through a web interface. Cloud infrastructure also improves scalability, reliability, and accessibility for multiple users such as medical students and researchers.

7.5 Human Effort Cost

Human effort represents the time, expertise, and research activities required to design, implement, evaluate, and validate the proposed AI-assisted diagnostic system. Although the research is conducted as part of an academic project, certain operational costs are associated with development activities, data preparation, and field-related research tasks.

The development process includes several activities such as designing the patient screening questionnaire, implementing machine learning models for cardiac and gastric risk prediction, developing backend APIs and the web-based user interface, and conducting system testing and validation experiments.

In addition, some travel and operational expenses may occur during the research process when collecting domain knowledge, validating system requirements, or consulting with medical professionals or healthcare environments for better understanding of chest pain diagnosis workflows.

Activity	Description	Estimated Cost (LKR)
System Design	Designing questionnaire structure, risk scoring rules, and system architecture	4,000
AI Model Development	Developing and training machine learning models for cardiac and gastric risk prediction	5,000
Backend Development	Implementation of FastAPI backend services and database integration	3,000
Frontend Development	Development of the web-based dashboard interface using React	2,000
System Testing & Validation	Testing the system functionality and evaluating model performance	2,000
Research Travel	Transportation expenses for meetings, consultations, or field observations related to the study	2,000
Documentation & Reporting	Preparation of research reports, system documentation, and proposal writing	2,000

Table 8: Human Effort and Research Activity Cost Estimation

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown per Member

Task ID	Task Name	Description	Timeline
T1	Literature Review	Review research papers related to chest pain diagnosis, cardiovascular risk prediction, gastritis symptoms, and machine learning applications in healthcare.	March 2026
T2	Questionnaire Design	Design a patient screening questionnaire to collect demographic information, symptoms, lifestyle factors, and medical history required for risk analysis.	March 2026
T3	Clinical Risk Scoring Logic Development	Develop rule-based scoring logic for calculating CardiacScore and GastricScore based on symptom characteristics and clinical risk factors.	March – April 2026
T4	Dataset Collection	Collect structured healthcare datasets related to cardiac risk factors and gastrointestinal symptoms from public medical data repositories.	April 2026
T5	Dataset Preparation	Clean, preprocess, and organize collected datasets to prepare them for machine learning model training and evaluation.	April – May 2026
T6	Feature Engineering	Convert questionnaire responses and patient attributes into structured features suitable for machine learning analysis.	May 2026
T7	Machine Learning Model Development	Develop machine learning models such as XGBoost and LightGBM to predict cardiac risk probability and gastric symptom probability.	May – June 2026
T8	Troponin Interpretation Module	Implement logic to analyze Troponin biomarker values and incorporate them into cardiac risk prediction.	June 2026

T9	Backend API Development	Develop FastAPI backend services to process questionnaire inputs, run risk prediction models, and return screening results.	July 2026
T10	System Integration	Integrate the baseline risk assessment component with the central system architecture and database.	July – August 2026
T11	Explainable AI Implementation	Implement explainable AI techniques (e.g., SHAP) to highlight key symptoms and risk factors influencing predictions.	August 2026
T12	System Testing and Validation	Evaluate the model performance using metrics such as accuracy, precision, recall, and ROC-AUC.	September 2026
T13	Performance Optimization	Improve model performance, refine scoring rules, and optimize system response time.	September – October 2026
T14	Documentation and Report Preparation	Prepare research documentation, system design reports, and final research thesis content.	October 2026

Table 9: Work Breakdown Structure (WBS) – Task Breakdown for Component 1 Development

8.2 Timeline Alignment

The development tasks for **Component 1 – Baseline Risk & Patient Profile Analysis** are aligned with the research project timeline to ensure that model development, integration, and evaluation are completed before the final submission deadline

Phase	Period	Activities
Phase 1	March 2026	Literature review and questionnaire design
Phase 2	March – April 2026	Development of clinical scoring logic and dataset collection
Phase 3	April – May 2026	Dataset preparation and feature engineering
Phase 4	May – June 2026	Machine learning model development and troponin analysis
Phase 5	July – August 2026	Backend API development and system integration

Phase 6	September – October 2026	System testing, evaluation, and optimization
---------	--------------------------	--

Table 10: Project Development Timeline for Component 1 – Baseline Risk & Patient Profile Analysis

8.3 Balanced Workload Distribution

The workload for Component 1 – Baseline Risk & Patient Profile Analysis is structured across multiple development stages to ensure balanced progress throughout the research project.

The initial stage focuses on research and system design, including literature review and development of the patient screening questionnaire. This stage ensures that relevant clinical indicators and symptom characteristics are properly captured for early risk assessment.

The second stage focuses on dataset preparation and feature engineering. During this stage, healthcare datasets are collected and preprocessed to train machine learning models capable of predicting cardiac and gastric risk probabilities.

The third stage involves machine learning model development and biomarker analysis. Algorithms such as XGBoost and LightGBM are implemented to analyze patient attributes and generate dual risk scores. Troponin biomarker interpretation is also incorporated into the risk assessment process.

The fourth stage focuses on system development and integration. Backend APIs are implemented using FastAPI, and the baseline risk assessment module is integrated with the central diagnostic system and database.

The final stage involves system testing, evaluation, and optimization to ensure that the component produces accurate predictions and operates efficiently within the overall AI-assisted diagnostic workflow.

This structured development approach ensures systematic progress from research and data preparation to AI model development, system integration, and final evaluation.

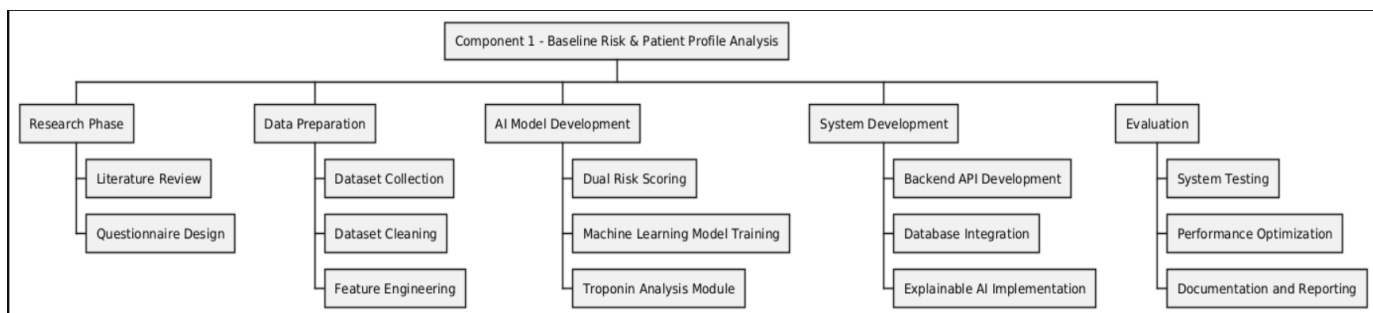


Figure 5: Hierarchical Work Breakdown Structure for Component 1 – Baseline Risk & Patient Profile Analysis

9. GANTT CHART

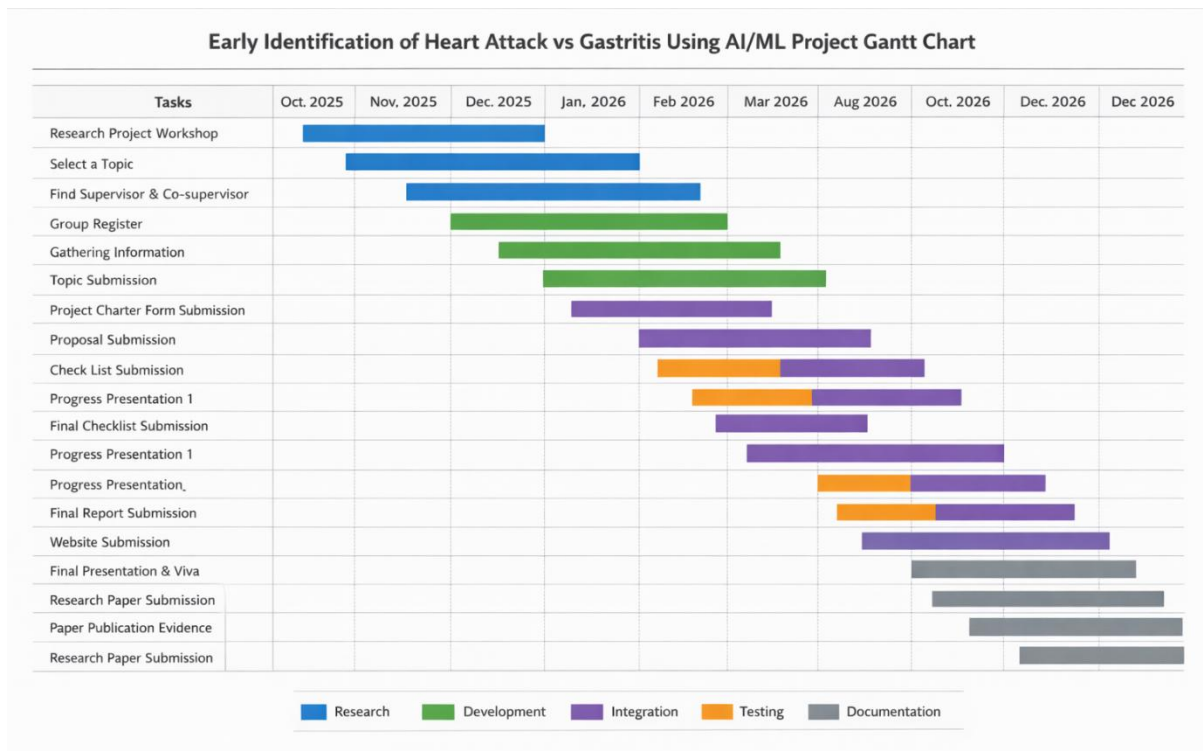


Figure 6: Gantt chart

References

- [1] N. Pervez, M. Afzal, F. U. Rehman, A. G. Khan, M. Hashim, S. A. Bari, R. Kumar and S. U. Kakar, "Comparison of Misconception of Heart Attack Pain with Heartburn: How Common Is It," *Indus Journal of Bioscience Research*, p. 203–208, January 2025.
- [2] A. Pushmika, Y. Wijesekara, T. Naragala, V. Muthukudaarachchi, P. K. W. Abeygunawardhana and R. Liyanage, "Predictive Analytics for Blood Supply Chain Management and Data Security in Healthcare System," *Proceedings of the IEEE International Conference on Advancements in Computing (ICAC)*, p. 1–6, December 2023.
- [3] M. Alam, S. Rahman and N. Chowdhury, "Mobile Health Intervention System in Optimizing Healthcare Delivery for Cardiac Patients," *Journal of Mobile Health Systems*, p. 45–52, March 2024.
- [4] P. Sharma and R. Gupta, "Atypical Myocardial Infarction Masquerading as Gastroenteritis," *Clinical Cardiology Case Reports*, p. 112–116, July 2023.
- [5] H. Sato, K. Dote and T. Uchida, "Takotsubo Cardiomyopathy Caused by Gastritis: A Clinical Case Study," *Journal of Cardiology Research*, p. 89–95, October 2022.

- [6] M. Rahman, T. Islam and S. Ahmed, "Blockchain-Based Decentralized Breast Cancer Patient Monitoring System for Remote Healthcare Security," *IEEE International Conference on Healthcare Informatics*, p. 321–326, November 2023.
- [7] M. Fernando, R. Silva and N. Perera, "VascularVision: Mixed Reality-Based Teaching and Learning Environment for Cardiovascular Conditions," *IEEE International Conference on Virtual Reality and Visualization*, p. 210–216, September 2023.
- [8] A. U. Haq, J. P. Li, M. H. Memon, S. Nazir and R. Sun, "A Hybrid Intelligent System Framework for the Prediction of Heart Disease Using Machine Learning Algorithms," *IEEE Access*, p. 355–365, June 2018.
- [9] U. R. Acharya, H. Fujita, S. L. Oh, Y. Hagiwara, J. H. Tan and M. Adam, "Deep Convolutional Neural Network for the Automated Detection of Myocardial Infarction Using ECG Signals," *IEEE Transactions on Biomedical Engineering*, p. 1–10, December 2017.
- [10] S. Rajendran and R. Iyer, "Machine Learning-Based Clinical Decision Support System for Cardiovascular Disease Prediction," *IEEE Journal of Biomedical and Health Informatics*, p. 1–8, April 2020.
- [11] A. Johnson, T. Pollard and L. Shen, "Machine Learning for Early Detection of Acute Coronary Syndrome," *IEEE Journal of Translational Engineering in Health and Medicine*, p. 1–9, September 2018.
- [12] P. K. Shillan, J. Sterne, J. Champneys and B. Gibbison, "Use of Machine Learning to Analyse Medical Data for Prediction of Cardiac Events," *IEEE Access*, p. 142–150, May 2019.
- [13] S. D. Nandhini and R. Radha, "An Intelligent Clinical Decision Support System for Heart Disease Prediction Using Data Mining Techniques," *IEEE International Conference on Smart Systems and Inventive Technology*, p. 1–6, January 2019.
- [14] R. K. Pathak, S. Tiwari and A. Mishra, "Heart Disease Prediction Using Data Mining Techniques," *IEEE International Conference on Computational Intelligence and Communication Technology*, p. 1–5, February 2018.
- [15] T. Weng, S. Li and Y. Wang, "Machine Learning Approaches for Chest Pain Risk Stratification in Emergency Departments," *IEEE Journal of Biomedical and Health Informatics*, p. 1–7, July 2020.
- [16] H. Chen, Z. Huang and W. Lin, "Artificial Intelligence in Cardiovascular Medicine: A Review of Current Applications," *IEEE Reviews in Biomedical Engineering*, p. 45–60, March 2021.

- [17] A. Esteva, B. Kuprel and R. Novoa, "A Guide to Deep Learning in Healthcare," *Nature Medicine*, p. 24–29, January 2019.
- [18] K. Shameer, K. Johnson, B. Glicksberg, J. Dudley and P. Sengupta, "Machine Learning in Cardiovascular Medicine," *IEEE Reviews in Biomedical Engineering*, p. 1–12, August 2018.
- [19] J. Luo, M. Wu, D. Gopukumar and Y. Zhao, "Big Data Application in Biomedical Research and Healthcare," *IEEE Access*, p. 913–920, October 2016.
- [20] E. Topol, "High-Performance Medicine: The Convergence of Human and Artificial Intelligence," *Nature Medicine*, p. 44–56, February 2019.
- [21] C. Krittanawong, H. Zhang, Z. Wang, M. Aydar and T. Kitai, "Artificial Intelligence in Precision Cardiovascular Medicine," *European Heart Journal*, p. 3277–3285, November 2017.
- [22] G. D. Clifford, C. Liu and B. Moody, "Advanced Methods and Tools for ECG Data Analysis," *IEEE Reviews in Biomedical Engineering*, p. 1–21, June 2016.

11. APPENDICES

Appendix A – Survey Instruments

This appendix contains the questionnaire used to collect patient symptom data related to chest pain and gastric conditions. The survey includes questions regarding patient demographics, lifestyle factors, meal-related symptoms, medication history, and other relevant clinical indicators that help differentiate cardiac and gastric chest pain.

Appendix B – Preliminary UI Mockups

This appendix includes preliminary user interface designs for the medical student dashboard. The mockups demonstrate how users can upload gastroscopy images, review AI predictions, view severity levels, and access treatment recommendations generated by the system.

Appendix C – Risk Analysis

Potential Risk	Description	Mitigation Strategy
Limited Dataset Availability	Insufficient endoscopy images for training the AI model may affect model performance.	Use publicly available datasets such as HyperKvasir and apply data augmentation techniques.

Model Accuracy Issues	The AI model may produce incorrect predictions due to limited or imbalanced data.	Use validation techniques, improve dataset quality, and tune model parameters.
System Integration Problems	Difficulty integrating the gastric analysis module with other system components.	Use modular architecture and well-defined APIs for communication between components.
Data Privacy Concerns	Handling medical images and patient data may raise privacy issues.	Ensure all datasets are anonymized and implement secure data storage methods.
Hardware or Resource Limitations	Limited computing power may slow down model training and testing.	Use cloud-based computing resources and optimize model training processes.

Appendix D – Ethical Documents

This appendix includes ethical considerations related to the use of medical datasets and patient information. All patient data used in the research will be anonymized to protect privacy, and datasets will be used strictly for research and educational purposes in accordance with medical data protection guidelines.